

Real analysis

1. Sets, relations, functions, and order

1.1. Logic

Theorem 1.1.1 (De Morgan's Laws for statements)

AND and OR interchange under complementation.

In C notation:

$$!(A \ || \ B) == (!A) \ \&\& \ (!B)$$

$$!(A \ \&\& \ B) == (!A) \ || \ (!B)$$

In mathematical logical notation:

$$\neg(A \vee B) = (\neg A) \wedge (\neg B)$$

$$\neg(A \wedge B) = (\neg A) \vee (\neg B)$$

Theorem 1.1.2 (De Morgan's Laws for set theory)

Union and intersection interchange under complementation.

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Theorem 1.1.3

Given the true statement $P \implies Q$,

- $\neg P \implies \neg Q$ is the **inverse**, which might be true or false.
- $Q \implies P$ is the **converse**, which is true iff the inverse is true.
- $\neg(P \implies Q)$ is the **negation**, which is always false.
- $\neg Q \implies \neg P$ is the **contrapositive**, which is always true.

1.2. Sets

See 04 Set Theory, [Topology] 01 Set Theory

A set is defined by selecting elements from another set such that $S(x)$ is true:

$$\{x \in A : S(x)\} \text{ or } \{x \in A \mid S(x)\}$$

$\emptyset$ denotes the empty set.

$P \implies Q$ means P implies Q (if P , then Q).

$P \Leftrightarrow Q$ means P and Q are equivalent (P iff Q).

$A \subset B$ means A is a subset of B (**$A = B$ is allowed**).

$A \subsetneq B$ means A is a strict/proper subset of B .

$A = B$ means $A \subset B$ and $B \supset A$.

$A \cup B$ means the union of A and B .

$A \cap B$ means the intersection of A and B .

$A \parallel B$ means $A \cap B = \emptyset$.

$A \setminus B = \{x \in A : x \notin B\}$, everything in A that's not in B .

$A \times B = \{(a, b) : a \in A, b \in B\}$ is called the **Cartesian product** of A and B .

Definition 1.2.1 (Power set)

The set of all subsets of A , denoted $\mathcal{P}(A)$.

1.3. Ordered pair

See 04 Set Theory

Definition 1.3.1 (Ordered pair)

The **ordered pair** (a, b) is the set $\{\{a\}, \{a, b\}\}$.

Theorem 1.3.2

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

1.4. Relations

See 04 Set Theory, [Topology] 02 Relations

Definition 1.4.1 (Relation)

A **relation** $R : A \rightarrow B$ is any subset of $A \times B$.

aRb means $(a, b) \in R$.

Theorem 1.4.2 (Relations can compose)

If $R : A \rightarrow B$ and $S : B \rightarrow C$, then $S \circ R : A \rightarrow C$ is a subset of $A \times C$.

Warning

$R \circ R^{-1}$ is not necessarily the identity map!

Theorem 1.4.3

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$$

$$(T \circ S) \circ R = T \circ (S \circ R)$$

Definition 1.4.4

R is **reflexive** iff aRa for all $a \in A$.

R is **symmetric** iff $aRb \Leftrightarrow bRa$.

R is **transitive** iff $aRb, bRc \Rightarrow aRc$.

1.4.1. Equivalence relations

Definition 1.4.5 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Definition 1.4.6 (Equivalence class)

Let $R : A \rightarrow B$ be an equivalence relation. Then the **equivalence class** of $a \in A$ is

$$[a] = [a]_R := \{b \in A : aRb\}$$

The set of all equivalence relations of R in A is called A/R .

Lemma 1.4.7

Two equivalence classes are either identical or disjoint.

Definition 1.4.8 (Partition)

A collection of disjoint nonempty subsets of A whose union is all of A .

Theorem 1.4.9

Given an equivalence relation on nonempty A , its set of equivalence classes forms a partition of A .

Given a partition P of A , there is a unique equivalence relation whose set of equivalence classes forms P .

1.4.2. Functions (maps)

See 03 Functions

Definition 1.4.10 (Function)

A **function** (also called a **map** or **mapping**) is a relation $f : A \rightarrow B$ where each $a \in A$ appears as the first element (input) of an ordered pair $(a, b) = afb$ exactly once.

If $(a, b) \in f$, we can also say $f(a) = b$. b is the **value** of f at a , or the **image** of a under f .

A is the **domain** and B the **codomain** of f .

Definition 1.4.11 (Injective)

A function f is **injective** or **one-to-one** iff:

- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 1.4.12 (Surjective)

A function $f : A \rightarrow B$ is **surjective** or **onto** iff:

- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Definition 1.4.13 (Bijective)

A function is **bijective** and has **one-to-one correspondence** if it is both injective and surjective.

Theorem 1.4.14

A composition of two functions is a function. If both are sur/in/bijective, so is their composition.

Definition 1.4.15

The set of all functions $A \rightarrow B$ is denoted B^A .

Lemma 1.4.16

If A and B are finite, then the size of B^A is $|B|^{|A|}$, the size of B to the size of A .

Theorem 1.4.17

For any set A , there is a bijection between $\mathcal{P}(A)$ and 2^A .

Definition 1.4.18 (Induced map)

The **induced map** of the relation $f : A \rightarrow B$ is a function $\bar{f} : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ defined as

$$\bar{f}(X) := \{b \in B : b = f(a) \text{ for some } a \in X\}$$

$\bar{f}(X)$ is called the **image** of X . This is sometimes denoted $f(X)$ (which is an abuse of notation).

$\bar{f}$ is always a function regardless of if f is a function.

1.5. Sequence

Definition 1.5.1 (Sequence)

A **sequence** is a function whose domain is the natural numbers, or some subset of the natural numbers.

If a is a sequence, $a(n)$ is also denoted a_n , and is called the n th element of a .

1.6. Cardinality

See 06 Counting and Sizes of Sets

Assume that there is some “overset” containing all of the sets we consider.

Definition 1.6.1 (Cardinality)

$\text{card}(A)$ is the equivalence class of A under bijection.

Equivalently, $\text{card}(A) = \text{card}(B)$ iff there is a bijection $A \rightarrow B$.

Two sets are **equipotent**, **equinumerous**, **equipollent**, or **similar** if they have the same cardinality.

Definition 1.6.2

$\text{card } A \leq \text{card } B$ iff there is an injection from A to B .

Theorem 1.6.3 (Schroeder-Bernstein Theorem)

$$\begin{aligned} \text{card}(A) \leq \text{card } B \text{ AND } \text{card}(B) \leq \text{card}(A) \\ \implies \text{card}(A) = \text{card}(B) \end{aligned}$$

Theorem 1.6.4 (Cantor's Theorem)

For any set A ,

$$\text{card}(A) < \text{card}(\mathcal{P}(A))$$

Theorem 1.6.5

Every infinite set is similar to a proper subset of itself.

No finite set is similar to a proper subset of itself.

1.7. Countability

Definition 1.7.1 (Countability)

S is countable iff there is an injection $f : S \rightarrow \mathbb{Z}^+$, or equivalently iff $\text{card } S \leq \text{card } \mathbb{Z}^+$.

Corollary 1.7.1.1

1. Every subset of a countable set is countable.
 - The elements of a countable set can be indexed by the positive integers.
 - All finite sets are countable.
2. A countable union of countable sets is countable.

Theorem 1.7.2

The product of a finite number of countable sets is countable.

However, a product of an infinite number of countable sets is not countable.

Theorem 1.7.3

$$\text{card}(\mathbb{R}) = \text{card}(\mathcal{P}(\mathbb{Z}^+))$$

Thus, the reals are not countable.

1.8. Axiom of choice

Axiom 1.8.1 (Axiom of Choice)

Given a infinite collection of disjoint finite sets, it is possible to select one element from each of the sets.

Equivalently, if A is a set of nonempty sets, there is always a function $f : A \rightarrow \bigcup A$ such that $f(a) \in a$ for all $a \in A$.

In this class, we assume the Axiom of Choice.

Theorem 1.8.2 (Zermelo's Theorem)

The following theorems are equivalent:

- the Axiom of Choice
- the Well-Ordering Property
- the Hausdorff Maximum Principle
- Zorn's Lemma

Theorem 1.8.3 (Well-Ordering Property (Zermelo's Postulate))

Every set can be well-ordered.

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Theorem 1.8.4 (Hausdorff Maximum Principle)

Every poset has a “maximal” totally ordered subset, i.e. a totally ordered subset that cannot be extended without escaping the poset.

Theorem 1.8.5 (Zorn's Lemma)

If A is a poset in which every totally ordered subset has an upper bound, then A has a maximal element, i.e. there is no larger element in the poset.

2. Fields and order

2.1. Equality

Definition 2.1.1 (Equality)

Equality is an equivalence relation on the reals, so:

- **Reflexive:** $x = x$.
- **Symmetric:** $x = y \implies y = x$.
- **Transitive:** $x = y$ and $y = z \implies x = z$.

2.2. Fields

Definition 2.2.1 (Field)

A *field* $\mathbb{F}$ is any set equipped with two relations, **addition** (+) and **multiplication** (*), which satisfy the following properties:

- **Closure axiom:** If x and y are in $\mathbb{F}$, so are both $x + y$ and xy .
- **Commutativity axioms:** $x + y = y + x$ and $xy = yx$.
- **Associativity axioms:** $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$.
- **Distributivity axiom:** $x(y + z) = xy + xz$.
- **Substitution axioms:** $w = x$ and $y + z \implies w + y = x + z$ and $wy = xz$.
- The **additive inverse axiom** and **multiplicative inverse axiom**, defined below.

Axiom 2.2.2 (Additive inverse axiom)

Given $x, y \in \mathbb{F}$, there exists $z \in \mathbb{F}$ s.t. $x + z = y$ (i.e. all addition problems have solutions).

We denote this z by the expression $(y - x)$. We also denote, for every x :

$$0_x := (x - x)$$

$$-x := (0_x - x)$$

Theorem 2.2.3 (Zero is unique)

There is some $0 \in \mathbb{F}$ s.t. $0 = 0_x = 0_y$ for all x and y in $\mathbb{F}$.

Axiom 2.2.4 (Multiplicative inverse axiom)

There exists some number in $\mathbb{F}$ which is not equal to 0. If $x, y \in \mathbb{F}$ s.t. $x \neq 0$, then there exists a real number z s.t. $xz = y$. This z is denoted (y/x) .

The number (x/x) is denoted 1_x , and $(1_x/x)$ is denoted x^{-1} and called the **reciprocal** of x .

Theorem 2.2.5 (One is unique)

There is some $1 \in \mathbb{F}$ s.t. for all nonzero x and y , $1 = 1_x = 1_y$.

Lemma 2.2.6

In any field $\mathbb{F}$, the following hold:

- $0x = 0$ for any x
- $0/x = 0$ for any nonzero x
- $-(-x) = x$
- $(x^{-1})^{-1} = x$ for any nonzero x
- $-(x - y) = y - x$
- $x - y = x + (-y)$
- $-(x + y) = (-x) + (-y)$
- $-x = -1x$

2.3. Order

Definition 2.3.1 (Order)

Let S be a set. An **order** $<$ is a relation that satisfies the order axioms:

- **Trichotomy:** For any $x, y \in S$, exactly one of:
 - $x < y$
 - $x = y$
 - $x > y$
- **Transitivity:** If $x < y$ and $y < z$, then $x < z$.

Definition 2.3.2 (Upper bound)

u is an **upper bound** for A iff $a \leq u$ for all $a \in A$.

Definition 2.3.3 (Supremum)

The **supremum** or **least upper bound** of A , denoted $\sup A$, is the unique object s.t. for every upper bound u of A , $\sup A \leq u$.

Definition 2.3.4 (Lower bound)

l is an **lower bound** for A iff $l \leq a$ for all $a \in A$.

Definition 2.3.5 (Infimum)

The **infimum** or **greatest lower bound** of A , denoted $\inf A$, is the unique object s.t. for every upper bound l of A , $\sup A \leq l$.

Theorem 2.3.6

If a set has a maximum, the maximum is its supremum.

If a set has a minimum, the minimum is its infimum.

Definition 2.3.7 (Dedekind completeness)

An ordered set is **complete**, **Dedekind-complete**, or has the **least upper bound property** iff:

- every subset of the set that has an upper bound has a supremum, or equivalently
- every subset of the set that has a lower bound has an infimum.

2.4. Ordered fields

Definition 2.4.1 (Ordered field)

A field $\mathbb{F}$ is an **ordered field** iff it has an order $<$ **compatible with arithmetic**, i.e. satisfying the properties:

- **Translation:** If $x < y$, then $x + z < y + z$ for all z .
- **Positive closure:** If $x > 0$ and $y > 0$, then $xy > 0$.

An element x is **positive** if $x > 0$ and **negative** if $x < 0$. The set containing all positive elements of $\mathbb{F}$ is denoted $\mathbb{F}^+$, and the set containing all negative elements of $\mathbb{F}$ is denoted $\mathbb{F}^-$.

Lemma 2.4.2

- $x > 0$ and $y > 0 \implies x + y > 0$
- $x > 0 \implies -x < 0$
- $-x > 0 \implies x < 0$

- 1 is positive
- $x > y$ and z is positive $\implies xz > yz$
- $x > y > 0 \implies 0 < x^{-1} < y^{-1}$

Theorem 2.4.3

If $a, b \in \mathbb{F}$ and $a \leq b + \varepsilon$ for every positive ε , then $a \leq b$ (if $\mathbb{F}$ is an ordered field).

2.5. Real numbers

Theorem 2.5.1

There is a unique complete ordered field, called $\mathbb{R}$, the **real numbers**.

Theorem 2.5.2 (Approximation Property)

Let S be a nonempty set of reals with a supremum. Then for every real number $a < \sup S$ there exists $x \in S$ with $a < x \leq \sup S$.

Theorem 2.5.3 (Additive Property)

Let A and B be sets of reals with suprema. Consider

$$C := \{a + b : a \in A, b \in B\}.$$

Then,

$$\sup C = \sup A + \sup B.$$

Theorem 2.5.4 (Comparison Property)

Let S and T be two nonempty sets of reals where $s \leq t$ for all $s \in S, t \in T$. Then, if T has a supremum, S also has a supremum and $\sup S \leq \sup T$.

Lemma 2.5.5

The positive integers have no upper bound.

Proof. Assume that $z = \sup \mathbb{Z}^+$ exists. We know that $z - 1 < z$, so by the approximation property there is

a positive integer s with $z - 1 < s$. Thus $z < s + 1$. $\square$

Definition 2.5.6 (Comensural)

The set X is **comensural** iff for any $x, y \in X$ with $x > 0$, then there is a positive integer n s.t. $nx > y$.

Theorem 2.5.7 (Archimedean property)

The reals are comensural.

Proof. Consider $\frac{y}{x}$. If $\frac{y}{x} < 0$, then $n = 0$. Otherwise, $\frac{y}{x} > 0$. Since the positive integers are unbounded, choose $n > \frac{y}{x}$. Then $nx > y$. $\square$

Theorem 2.5.8 (Density of the rationals / rationals between reals)

Between any two reals there is a rational number.

Proof. The other cases are trivial, so consider $0 < x < y$. Choose an integer $n > \frac{1}{y-x}$.

We define $q := \frac{k}{n}$, and

We claim $x < q$. Then:

$$\frac{1}{y-x} < n$$

$$y-x > \frac{1}{n}$$

$$y - \frac{1}{n} > x$$

$$y - q \leq \frac{1}{n} < y - x$$

2.6. Extended reals

Definition 2.6.1 (Extended reals)

The set $\mathbb{R}^*$ consisting of the reals, plus two elements $-\infty$ and $+\infty$.

The order on it is defined such that

$$-\infty < x < +\infty \text{ for all } x \in \mathbb{R}$$

and when comparing two reals it still has the standard order.

All expressions whose value is defined follow the rules of complete ordered fields, but there are many undefined expressions.

2.7. Absolute values

Theorem 2.7.1 (Corollaries of the Triangle Inequality)

For any $x, y \in \mathbb{R}$:

$$|x + y| \leq |x| + |y|$$

$$|x - y| \leq |x| + |y|$$

$$||x| + |y|| \leq |x - y|$$

2.8. Bounded function

Definition 2.8.1 (Bounded function)

$f : S \rightarrow \mathbb{R}$ is **bounded** iff the image $f(S)$ has both a supremum and infimum.

Equivalently, f is bounded iff there exists M s.t. $|f(x)| < M$ for all x .

Equivalently, f is bounded iff $f(S)$ is bounded.

Theorem 2.8.2

If $f(x) \leq g(x)$ for all $x \in D$, then $\sup f(D) \leq \sup g(D)$.

2.9. Decimal representation

Definition 2.9.1 (Decimal representation)

A string of the form

$$(-)d_n d_{n-1} \dots d_1 d_0 . d_{-1} d_{-2} \dots$$

where each $d_i \in \{0, 1, 2, \dots, 9\}$.

i.e.

- optionally, a minus sign
- a finite string of digits
- a decimal point
- an infinite string of digits

Theorem 2.9.2

There is a bijection between $\mathbb{R}^+$ and the set of decimal representations without a minus sign and such that there is a last 9, i.e. the set of nonterminating positive decimal representations.

In order to prove this, we need some lemmas:

Lemma 2.9.3

There is a bijection between $\mathbb{Z}^+$ and the set of decimal representations ending in an infinite string of 0s.

Proof. Every integer is representable as

$$n = 10m + r.$$

So keep dividing the integer by 10, and pushing the remainder onto the front of the string. This gives us a base-10 representation of the integer, i.e. a finite string of digits.

To make a full decimal representation, put an infinite string of 0s after the decimal point. $\square$

Now let's prove the original theorem.

Proof. Let $x \in \mathbb{R}^+$. Take $n = \lfloor x \rfloor$ and find its decimal representation. Then $0 < x - n < 1$. $\square$

Now we need to prove more lemmas:

Lemma 2.9.4

Every decimal representation represents a real number.

3. Sequences

Definition 3.1 (Sequence)

A function $x : \mathbb{Z}^+ \rightarrow \mathbb{R}$.

$x(n)$ is usually denoted x_n and called the n th element of the sequence. The sequence itself is often denoted $\{x_n\}$, $\{x_n\}_{n=1}^\infty$, (x_n) , or similar.

If the terms of the sequence are specified by a formula, the sequence might be left unnamed and written as, for example, $\{\frac{1}{n}\}_{n=1}^\infty$.

Definition 3.2 (Limit of a sequence)

Let $\{x_n\}$ be a sequence. L is the **limit**, denoted $\lim_{n \rightarrow \infty} x_n$, of the sequence iff for every $\varepsilon > 0$ there exists N s.t. whenever $n > N$, $|x_n - L| < \varepsilon$. If such a limit exists, the sequence **converges** to that limit.

A convergent sequence has exactly one limit.

If a sequence has no limit, it **diverges**.

Theorem 3.3

If a sequence is unbounded, it diverges.

Definition 3.4 (Monotonic)

Let $\{x_n\}$ be a sequence. If $x_n \leq x_{n+1}$ for all n , the sequence is **non-decreasing** or **increasing**. If $x_n < x_{n+1}$ for all N , the sequence is **strictly increasing**. If $x_n \geq x_{n+1}$ for all n , the sequence is **non-increasing** or **decreasing**. If $x_n > x_{n+1}$ for all N , the sequence is **strictly decreasing**.

In all these cases, the sequence is **monotonic** or **monotone**, and if it is strictly increasing or strictly decreasing, then the sequence is **strictly monotonic**.

Theorem 3.5

A bounded monotonically increasing sequence converges to its supremum.

A bounded monotonically decreasing sequence converges to its infimum.

Theorem 3.6

Every nonempty bounded subset of the reals has an increasing sequence converging to its supremum and a decreasing sequence converging to its infimum.

3.1. Subsequences

Definition 3.1.1 (Subsequence)

If $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is a strictly increasing function and $\{x_n\}$ is a sequence, $\{x_{k(n)}\}$ is a **subsequence** of $\{x_n\}$.

Theorem 3.1.2

The following are equivalent:

1. The sequence $\{x_n\}$ converges.
2. The t -tail $\{x_{n+t}\}$ converges for every t .
3. There is some t for which the t -tail converges.

3.2. Limit properties and theorems

Theorem 3.2.1

Let $\{x_n\} \rightarrow x$ and $\{y_n\} \rightarrow y$, s.t. $x_n \leq y_n$ for all n . Then, $x \leq y$.

Theorem 3.2.2 (Squeeze Theorem)

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences s.t. $a_n \leq b_n \leq c_n$ for all n and $a_n \rightarrow L$ and $c_n \rightarrow L$. Then $b_n \rightarrow L$.

Proof. Let $\varepsilon > 0$. Choose n so large that both $|a_n - L|, |c_n - L| < \varepsilon$. Then,

$$-\varepsilon < c_n - L < \varepsilon$$

$$-\varepsilon < a_n - L < \varepsilon$$

and thus

$$-\varepsilon < a_n - L \leq b_n - L \leq c_n - L < \varepsilon$$

so clearly

$$|b_n - L| < \varepsilon.$$

Since this works for all ε , $b_n \rightarrow L$. □

Theorem 3.2.3

Let $\{x_n\} \rightarrow x$ and $\{y_n\} \rightarrow y$. Then

1. $\{x_n + y_n\} \rightarrow x + y$.
2. $\{x_n - y_n\} \rightarrow x - y$.
3. $\{x_n y_n\} \rightarrow xy$.
4. $\{x_n / y_n\} \rightarrow x / y$ if $y_n \neq 0$ and $y \neq 0$ for all n .

Theorem 3.2.4

Let $\{x_n\}$ be a sequence and x a number. Then,

$$\{x_n\} \rightarrow x \iff \{x_n - x\} \rightarrow 0.$$

Theorem 3.2.5 (Convergence of geometric series)

Let $c > 0$. Then,

1. $c < 1 \implies \lim_{n \rightarrow \infty} c^n = 0$.
2. $c > 1 \implies \{c^n\}$ is unbounded.

Theorem 3.2.6 (Ratio Test)

Let $\{x_n\}$ be an always-nonzero sequence s.t.

$$\lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|} = L.$$

Then,

- $L < 1 \implies \{x_n\} \rightarrow 0$.
- $L > 1 \implies \{x_n\}$ is unbounded.

Theorem 3.2.7

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

3.3. Limsup and liminf

Definition 3.3.1 (Limsup and liminf)

Let $\{x_n\}$ be a bounded sequence. For each n , define

$$a_n := \sup\{x_k : k \geq n\}$$

$$b_n := \inf\{x_k : k \geq n\}$$

Thus, a_n is above the n -tail of the sequence, while b_n is below it.

Define

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} a_n$$

$$\liminf_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} b_n.$$

We know that $\limsup x_n \geq \liminf x_n$.

Theorem 3.3.2

If $\{x_n\}$ is bounded, then:

- there is a subsequence $\{x_{n_k}\} \rightarrow \limsup x_n$.
- there is a subsequence $\{x_{m_k}\} \rightarrow \liminf x_n$.

Theorem 3.3.3

Let $\{x_n\}$ be a bounded sequence with $\limsup x_n = \liminf x_n = L$. Then, $\{x_n\} \rightarrow L$.

Theorem 3.3.4

$\{x_n\} \rightarrow L$ iff every subsequence converges to L .

3.4. Bolzano-Weierstrass

Theorem 3.4.1 (Bolzano-Weierstrass for Sequences)

Every bounded sequence of real numbers (or in $\mathbb{R}^n$) has a convergent subsequence.

In topology terms, a subset of $\mathbb{R}^n$ is sequentially compact iff it is closed and bounded.

Definition 3.4.2 (Open ball)

Let $x \in \mathbb{R}^n$. The ball $B(x, \varepsilon)$ of radius ε about x is defined as

$$B(x, \varepsilon) := \{y \in \mathbb{R}^n : |x - y| < \varepsilon\}.$$

Definition 3.4.3 (Accumulation point)

Let S be a set. x is a **accumulation point** of S iff every ball $B(x, \varepsilon)$ contains a point of S other than x .

Theorem 3.4.4 (Bolzano-Weierstrass theorem)

Every bounded infinite subset of $\mathbb{R}^n$ has an accumulation point.

Theorem 3.4.5 (Cantor Intersection Theorem)

Let $S_1, S_2, \dots$ be a sequence of bounded closed subsets of $\mathbb{R}^n$ nested as follows: $S_1 \supset S_2 \supset \dots$. Then, the intersection $S_1 \cap S_2 \cap \dots$ is closed and nonempty.

3.5. Infinite limits

See 16 Infinite Limits

Definition 3.5.1 (Diverging to infinity)

A sequence **diverges to** $+\infty$ iff for any $M \in \mathbb{R}$ there is an N s.t. for all $n \geq N$ then $x_n \geq M$.

A sequence **diverges to** $-\infty$ iff for any $M \in \mathbb{R}$ there is an N s.t. for all $n \leq N$ then $x_n \leq M$.

Theorem 3.5.2

An unbounded increasing sequence diverges to $+\infty$.
An unbounded decreasing sequence diverges to $-\infty$.

Definition 3.5.3

Let $\{x_n\}$ be a sequence.

$\limsup x_n = +\infty$ iff $\{x_n\}$ diverges to $+\infty$.

$\liminf x_n = -\infty$ iff $\{x_n\}$ diverges to $-\infty$.

3.6. Cauchy sequences

Definition 3.6.1 (Cauchy sequence)

A sequence $\{x_n\}$ is called **Cauchy sequence** iff for every $\varepsilon > 0$, there exists N s.t. for all $m, n > N$, $|x_n - x_m| < \varepsilon$.

In short, beyond a certain term, all the following terms are arbitrarily close.

Theorem 3.6.2

Every Cauchy sequence is bounded.

Theorem 3.6.3

A sequence of reals converges iff it is a Cauchy sequence.

4. Series

4.1. Series

Definition 4.1.1 (Series)

Given a sequence $\{x_n\}$, the object

$$\sum_{n=1}^{\infty} x_n$$

is called an **infinite series**.

Given a series $\sum_{n=1}^{\infty} x_n$, define its k^{th} **partial sum** by

$$s_k = \sum_{n=1}^k x_n := x_1 + x_2 + \dots + x_n.$$

Since this sum is finite it is a real number. Additionally, we can recursively define this sum as

$$\begin{aligned}s_1 &:= x_1 \\ s_{k+1} &:= s_k + x_{k+1}.\end{aligned}$$

A series **converges** iff its sequence of partial sums converges, and it converges to the same value as its sequence of partial sums. It is considered to be equal to said value.

Series with different starting indices are defined in the obvious way:

$$\sum_{n=a}^{\infty} x_n := \sum_{n=1}^{\infty} x_{n+a-1}$$

4.2. Basic convergence tests

Theorem 4.2.1 (n^{th} term test for divergence)

If $\sum x_n$ converges, then $x_n \rightarrow 0$.

Equivalently, if $x_n \not\rightarrow 0$, $\sum x_n$ diverges.

Theorem 4.2.2

Let $\sum x_n = x$ and $\sum y_n = y$. Then, $\sum (x_n + y_n) = x + y$.

For any $c \in \mathbb{R}$, $\sum (cx_n) = cx$.

4.3. Absolute convergence

See 19 Absolute Convergence

Lemma 4.3.1

If $x_n \geq 0$ for all n , then $\sum x_n$ converges iff the sequence of partial sums is bounded above.

Furthermore, each partial sum is less than or equal to the sum.

Definition 4.3.2 (Absolute convergence)

Iff $\sum |x_n|$ converges, then $\sum x_n$ **absolutely converges**. If $\sum |x_n|$ diverges, but $\sum x_n$ converges, then $\sum x_n$ **conditionally converges**.

Theorem 4.3.3

If a series converges absolutely, it converges.

Theorem 4.3.4 (Comparison test for positive series)

Let $\sum x_n$ and $\sum y_n$ be series with $0 \leq x_n \leq y_n$ for all n . Then:

1. If $\sum y_n$ converges, then $\sum x_n \leq \sum y_n$ ($\sum x_n$ converges).
2. If $\sum x_n$ diverges, so does $\sum y_n$.

Corollary 4.3.4.1

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.

4.4. Ratio test and root test

Theorem 4.4.1 (Ratio test)

Let $\sum x_n$ be a series of nonzero terms. Suppose that

$$L := \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

Theorem 4.4.2 (Root test)

Let $\sum x_n$ be a series and $L := \limsup \sqrt[n]{|x_n|}$ (which may be infinite). Then,

- If $L < 1$, $\sum x_n$ converges absolutely.
- If $L > 1$, $\sum x_n$ diverges.
- If $L = 1$, the test gives no information.

4.5. Alternating series

See 20 Alternating Series

Theorem 4.5.1 (Alternating series test)

Let $x_n \rightarrow 0$ be monotone decreasing. Then, $\sum (-1)^n x_n$ converges.

Theorem 4.5.2

Let $\sum x_n$ be an absolutely convergent series converging to L . Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then, $\sum x_{\sigma n}$ is also absolutely convergent and converges to L .

In short, if the series is absolutely convergent, we can rearrange the terms and the sum is the same.

Theorem 4.5.3 (Alternating Series Rearrangement Theorem)

Let $\sum x_n$ be a series that satisfies the alternating series test but which is not absolutely convergent. Then, for any $L \in \mathbb{R}$ there is a rearrangement of the series that sums to L .

4.6. Multiplying series

Definition 4.6.1 (Cauchy product)

The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is defined as

$$\sum_{k=1}^n a_n b_{n+1-k}.$$

Essentially, we collect the terms of the product of $\sum a_n$ with $\sum b_n$ in diagonal strips when the terms are arranged in a grid.

Theorem 4.6.2 (Mertens' theorem)

Let $\sum a_n \rightarrow A$ and $\sum b_n \rightarrow B$. Then if at least one of these is absolutely convergent, then $\sum c_n \rightarrow AB$,

where $\sum c_n$ is the Cauchy product of $\sum a_n$ and $\sum b_n$.

Additionally, if both converge absolutely, so does the sum of c_n s.

4.7. Power series

Definition 4.7.1 (Power series)

Given $x_0 \in \mathbb{R}$ and a sequence a_n , we can create the **power series**, a function of x :

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

A power series **converges** iff there exists $x \neq x_0$ that makes the series converge. If $x = x_0$ makes the series converge, then the series converges for any x .

Theorem 4.7.2

A power series either diverges for all $x \neq x_0$, converges absolutely for all x , or there exists a real number ρ , called the **radius of convergence**, s.t. the series converges absolutely on $(x_0 - \rho, x_0 + \rho)$ and diverges if $|x - x_0| > \rho$.

When $x = x_0 + \rho$ or $x = x_0 - \rho$, this theorem is inconclusive.

5. Limits and continuity

5.1. Limits

Definition 5.1.1 (Accumulation point)

Let S be a set. x is a **accumulation point** of S iff every ball $B(x, \varepsilon)$ contains a point of S other than x .

Lemma 5.1.2

x is an accumulation point of S iff there is a sequence of points in $S - \{x\}$ converging to x .

Definition 5.1.3 (Limit)

Let S be a subset of $\mathbb{R}$ and c be an accumulation point of S . Let $f : S \rightarrow \mathbb{R}$ be a function.

The **limit** of $f(x)$ as $x \rightarrow c$ is L iff for each $\varepsilon > 0$, there exists a $\delta > 0$ s.t. whenever $x \in S - \{c\}$ and $|x - c| < \delta$, then $|f(x) - L| < \varepsilon$. If so, we say that $f(x)$ converges to L as $x \rightarrow c$.

If there is no such limit, then the limit does not exist and f **diverges**.

Limits are unique.

Theorem 5.1.4

Let S be a subset of $\mathbb{R}$ and c be an accumulation point of S . Let $f : S \rightarrow \mathbb{R}$ be a function.

Then $f(x) \rightarrow L$ as $x \rightarrow c$ iff for every sequence $\{x_n\}$ in $S - \{c\}$ whose limit is c , the sequence $\{f(x_n)\} \rightarrow L$.

5.2. Continuity

Definition 5.2.1 (Continuity)

Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \rightarrow \mathbb{R}$ be a function. Then, f is **continuous** at c iff for every $\varepsilon > 0$, there is a $\delta > 0$ s.t. whenever $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

Theorem 5.2.2

Let $S \in \mathbb{R}$ and $f : S \rightarrow \mathbb{R}$ be a function. Let c be a point in S . Then:

1. If c is not an accumulation point of S , then f is continuous at c .
2. If c is an accumulation point of S , then f is continuous at c iff $\lim_{x \rightarrow c} f(x) = f(c)$.
3. f is continuous at c iff for every sequence $\{x_n\} \in S$ with limit c , $f(x_n) \rightarrow f(c)$.

Theorem 5.2.3 (Composition of continuous functions is continuous)

Let $f : S \rightarrow \mathbb{R}$. If g is continuous at $c \in T$ and f is continuous at $g(c)$, then $f \circ g : T \rightarrow \mathbb{R}$ is continuous at c .

Theorem 5.2.4

There are no functions that are continuous at every rational but discontinuous at every irrational.

5.3. Big Continuity Theorems

Definition 5.3.1 (Bounded function)

A function $f : S \rightarrow \mathbb{R}$ is bounded iff its image is bounded.

Lemma 5.3.2

Let $[a, b]$ be a closed finite interval, and f continuous on $[a, b]$. Then f is bounded.

Theorem 5.3.3 (Extreme Value Theorem)

Let $f(x)$ be a continuous function defined on a closed interval $[a, b]$. Then, f achieves an absolute maximum and minimum on $[a, b]$.

Lemma 5.3.4

Let $f(x)$ be continuous on the closed bounded interval $[a, b]$. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in [a, b]$ s.t. $f(c) = 0$.

Theorem 5.3.5 (Intermediate Value Theorem)

Let f be continuous on closed, bounded $[a, b]$. Let $y \in \mathbb{R}$ s.t. $f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then, there exists $c \in (a, b)$ with $f(c) = y$.

5.4. Uniform continuity

Definition 5.4.1 (Uniform continuity)

Let $S \subset \mathbb{R}$ and $f : S \rightarrow \mathbb{R}$ be a function. For any $\varepsilon > 0$, f is **uniformly continuous** on S iff we can find δ s.t. for all $x, c \in S$ with $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

This differs from continuity iff the same δ works for every point in S .

Uniform continuity obviously implies continuity.

Theorem 5.4.2

If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function defined on a closed bounded interval, it is uniformly continuous.

Lemma 5.4.3

If $f : S \rightarrow \mathbb{R}$ is uniformly continuous and $\{x_n\}$ is a Cauchy sequence in S , then $\{f(x_n)\}$ is a Cauchy sequence.

Theorem 5.4.4

Let $f : (a, b) \rightarrow \mathbb{R}$. Then f is uniformly continuous iff it has a right-handed limit at a , a left-handed limit at b , and the extension of f to the closed interval $[a, b]$ is continuous.

5.5. Lipschitz continuity

Definition 5.5.1 (Lipschitz continuity)

Let $f : S \rightarrow \mathbb{R}$ be a function. If there exists $K \in \mathbb{R}$ s.t. for every $x, y \in S$, $|f(x) - f(y)| < K|x - y|$ then f is **Lipschitz continuous**.

Theorem 5.5.2

Lipschitz continuity implies uniform continuity.

6. Derivatives

Definition 6.1 (Derivative)

Let $f(x)$ be a real-valued function defined on an open interval (a, b) , and let $x_0 \in (a, b)$. The **derivative** of f at x_0 is

$$\frac{df}{dx}(x_0) := f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Iff this limit is defined, the function is **differentiable** at x_0 .

Theorem 6.2 (Continuity theorem)

Let f be defined on (a, b) with $x_0 \in (a, b)$. Then, f is differentiable at x_0 iff there is a function f^* (which depends on f and on x_0) s.t.

$$f(x) = f(x_0) + f^*(x)(x - x_0)$$

with f^* defined on (a, b) and continuous at x_0 , with $f^*(x_0) = f'(x_0)$.

If f is continuous, f^* can be defined as

$$f^*(x) := \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} & \text{if } x \neq x_0 \\ f'(x_0) & \text{if } x = x_0. \end{cases}$$

Definition 6.3 (One-sided derivative)

Let f be defined on $[a, b)$. The **right-hand derivative** of f at a is

$$f'_+(a) := \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}.$$

Let f be defined on $(a, b]$. The **left-hand derivative** of f at a is

$$f'_-(a) := \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}.$$

Definition 6.4 (Infinite derivatives)

Let f be continuous on $[a, b)$. The right-hand derivative of f at a is $+\infty$ iff

$$\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = +\infty.$$

The definition is similar for left-hand and negative infinite derivatives.

Theorem 6.5 (Sum, product, and quotient rules)

Let f and g be functions defined on an interval (a, b) , both differentiable at the point $x_0 \in (a, b)$. (In particular, the derivatives there cannot be infinite.) Then, the following derivatives exist and take on the following values:

- $(f \pm g)'(x_0) = f'(x_0) \pm g'(x_0)$.
- $(f * g)'(x_0) = f(x_0)g'(x_0) + f'(x_0)g(x_0)$.
- $(f/g)'(x_0) = \frac{g(x_0)f'(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}$.

Theorem 6.6 (Chain rule)

Let $f : (a, b) \rightarrow \mathbb{R}$ and $g : f(a, b) \rightarrow \mathbb{R}$. Let $x_0 \in (a, b)$ with f differentiable at x_0 and $f(x_0)$ contained in the interior of $f(a, b)$, and let g be differentiable at $f(x_0)$. Then, $(g \circ f)(x)$ is differentiable at x_0 , and

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0),$$

or equivalently,

$$\left. \frac{dg}{dx} \right|_{x_0} = \left. \frac{dg}{df} \right|_{f(x_0)} * \left. \frac{df}{dx} \right|_{x_0}.$$

6.1. Mean value theorem

Theorem 6.1.1

Let $f'(c) > 0$ and f continuous in some interval about c . Then, there is a $\delta > 0$ s.t.:

- $c < x < c + \delta \implies f(c) < f(x)$
- $c - \delta < x - c \implies f(x) < f(c)$

Equivalently, if the derivative is positive, then the function is increasing at that point.

Similarly, if the derivative is negative, then the function is increasing at that point.

Theorem 6.1.2 (Rolle’s theorem)

Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b) , s.t. $f(a) = f(b)$. Then, there exists $c \in (a, b)$ for which $f'(c) = 0$.

Theorem 6.1.3 (Mean value theorem)

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists some point $c \in (a, b)$ for which

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 6.1.4 (Cauchy’s mean value theorem)

Let $f(t)$ and $g(t)$ be two functions continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists some point $c \in (a, b)$ for which

$$f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a)).$$

Theorem 6.1.5

Let $f(x)$ be continuous on $[a, b]$ and differentiable on (a, b) . Then:

- $f'(x) > 0$ for all $x \in (a, b) \iff f$ is strictly increasing

- $f'(x) < 0$ for all $x \in (a, b) \iff f$ is strictly decreasing
- $f'(x) = 0$ for all $x \in (a, b) \iff f$ is constant

Let $g(x)$ also be continuous on $[a, b]$ and differentiable on (a, b) . Then:

- $g'(x) = f'(x)$ for all $x \in (a, b) \iff f(x) - g(x)$ is constant

Theorem 6.1.6

Let $f(x)$ be continuous on $(a, b]$ and differentiable on (a, b) , with $\lim_{x \rightarrow b^-} f'(x) = L$. Then, f is left-hand differentiable at b , and $f'_-(b) = L$.

A similar theorem exists for right-hand derivatives.

Theorem 6.1.7 (Darboux theorem)

Let $f(x)$ be a function that is differentiable on $[a, b]$. Let y be a value s.t. $f'(a) < y < f'(b)$ or $f'(a) > y > f'(b)$. Then there is some $c \in (a, b)$ s.t. $f'(c) = y$.

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